

$$2) \begin{cases} 2x_1 + 4x_2 = -4 \\ 5x_1 + 7x_2 = 11 \end{cases}$$

$$\text{Augmented matrix: } \left[\begin{array}{cc|c} 2 & 4 & -4 \\ 5 & 7 & 11 \end{array} \right]$$

$$\text{Row 1} = \text{Old row 1} \times \frac{1}{2}$$

$$\left[\begin{array}{cc|c} 1 & 2 & -2 \\ 5 & 7 & 11 \end{array} \right]$$

$$\text{Row 2} = \text{Row 2} - 5 \times \text{Row 1}$$

$$\left[\begin{array}{cc|c} 1 & 2 & -2 \\ 0 & -3 & 21 \end{array} \right]$$

$$\text{Row 2} = \text{Row 2} \times \left(-\frac{1}{3}\right) \rightarrow$$

$$\left[\begin{array}{cc|c} 1 & 2 & -2 \\ 0 & 1 & -7 \end{array} \right]$$

$$\text{Row 1} = \text{Row 1} + (-2) \times \text{Row 2}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 12 \\ 0 & 1 & -7 \end{array} \right]$$

$$\therefore x_1 = 12, x_2 = -7$$

3) Find the point (x_1, x_2) that lies on the line $x_1 + 5x_2 = 7$ and on the line $x_1 - 2x_2 = -2$.

$$\text{Augmented matrix: } \left[\begin{array}{cc|c} 1 & 5 & 7 \\ 1 & -2 & -2 \end{array} \right]$$

$$\text{Row 2} = \text{Row 2} - \text{Row 1}$$

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & -7 & -9 \end{array} \right]$$

$$\text{New row 2} = \text{Old row 2} \times \left(-\frac{1}{7}\right)$$

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 1 & 9/7 \end{array} \right]$$

$$\text{New row 1} = \text{Old row 1} + (-5) \times \text{row 2}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 4/7 \\ 0 & 1 & 9/7 \end{array} \right]$$

$$\therefore x_1 = 4/7, x_2 = 9/7$$

4) Find the point of intersection of the lines $x_1 - 5x_2 = 1$ and $3x_1 - 7x_2 = 5$.

$$\text{Augmented matrix: } \left[\begin{array}{cc|c} 1 & -5 & 1 \\ 3 & -7 & 5 \end{array} \right]$$

$$\text{Old row 2} - \text{New row 2} \times (-3)$$

$$\left[\begin{array}{cc|c} 1 & -5 & 1 \\ 0 & 8 & 2 \end{array} \right]$$

$$\text{New row 2} = \text{Old row 2} + (-3) \times \text{row 1}$$

$$\text{Old row 2} / 8 = \text{New row 2}$$

$$\left[\begin{array}{cc|c} 1 & -5 & 1 \\ 0 & 1 & 1/4 \end{array} \right]$$

$$\text{new row 1} = \text{old row 1} + 5 \times \text{row 2}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 9/4 \\ 0 & 1 & 1/4 \end{array} \right]$$

$$\therefore x_1 = 9/4, x_2 = 1/4$$